Biochemistry Cumulative Examination

October 11, 2013

1. What major metabolic pathway ends in pyruvic acid? 10 points

2. Draw the structure of pyruvic acid. 10 points

3. Pyruvic acid is sometimes converted to lactic acid. Under what conditions and why does this transformation take place? 15 points

4. What cofactor(s), if any, are required for this conversion? 15 points

5. Draw the structure of lactic acid. 10 points

6. In humans, what normally happens to the lactic acid that is produced? 10 points

7. In humans, when pyruvic acid (pyruvate) is not converted into lactic acid, what metabolic pathway does it enter? 10 points

8. What is pyruvate converted to for entry into this pathway? What is the name of the enzyme that carries it out and what cofactors are required? 20 points